

ACE–SCN / CSC / PhaseMirror-HQ Integration

Technical Report and Defensive Publication

Citizen Gardens
The Foundation of Multiplicity

April 25, 2026

Abstract

This document consolidates and formalizes recent developments around the Arithmetic Control Engine (ACE), the Structured Control Network (SCN), the Contraction Spectral Certificate (CSC), and their proposed unification with a PhaseMirror-HQ memory-kernel architecture. It is written as both a technical report and a prior-art defensive publication. The report covers: (i) executive-level motivation and system architecture; (ii) a comprehensive mathematical overview of ACE–SCN, including feasibility maps and certified bounds; (iii) the Track A / Track B split and the canonical 5,087-constraint zero-knowledge circuit layout; (iv) the proposal to treat an FZS-MK memory kernel and Zeno projection as the sole semantic authority for drift and protection metrics; (v) an ADR-centered governance plan with SWOT and RAID analysis; and (vi) implementation sketches and validation pathways sufficient to serve as public technical disclosure.

Contents

1	Executive Summary	3
2	Mathematical Overview of ACE–SCN	4
2.1	Arithmetic Objects and Hermitianization	4
2.2	Gap Functionals and Perturbation Classes	4
2.3	Projection and Feasibility Map	5
2.4	Certificates and Key Lemmas	6
2.5	SCN Amortized Controller	6
3	Governance Metrics and Zero-Knowledge Layout	7
3.1	Track A Governance Metrics	7
3.2	Track B Circuit and Constraint Budget	8
3.3	Commitment Strategy	8
4	PhaseMirror-HQ Unification and Kernel Authority	9
4.1	Motivation for Single Semantic Authority	9
4.2	Kernel Telemetry Contract	9
4.3	SCN Conditioning on Kernel Telemetry	10
5	ADR, SWOT, and RAID Structure	10
5.1	ADR Package	10
5.2	SWOT Analysis	11
5.3	RAID Log (Risks, Assumptions, Issues, Dependencies)	11
6	Code Snippets and Implementation Sketches	12
6.1	Python: Kernel Telemetry and Certificate Payload	12
6.2	Circom Witness Binding Sketch	13
7	Validation Path and Repository Changes	13
7.1	Four-Phase Validation Plan	13
7.2	Recommended Repository Edits	13
8	Defensive Publication Notes	14
9	Conclusion	14
A	Mathematical Appendix: Proofs and Operator Norm Bounds	15
A.1	Weyl-Type Bounds via Frobenius Norm	15
A.2	Feasibility-Map Invariants in All Modes	15
A.3	Near-Span Implies Near-Commuting	17
A.4	Monotonicity and Scaling Behaviour in Mode 3	18
A.5	Operator Norm Bounds in the Drift Quotient Constraint	19
A.6	Summary	20

1 Executive Summary

The Arithmetic Control Engine (ACE) is a reproducible system for amortized, certified spectral shaping on families of arithmetic operators, in particular self-adjoint Hecke operators on cusp forms.¹ Given a Hermitian operator matrix A and a target gap value $\hat{g}$, ACE produces a Hermitian perturbation Δ that satisfies hard feasibility certificates and, optionally, preserves arithmetic structure via explicit fidelity modes. The Structured Control Network (SCN) proposes such perturbations in an amortized fashion, while a deterministic feasibility map enforces admissible constraints independently of training.²

The implementation roadmap partitions ACE into two tracks. Track A (Python) computes semantic quantities including drift metrics, windowed contraction (WAC), and certification flags, while Track B (Rust/Circom) implements a fixed-topology zero-knowledge circuit that verifies algebraic invariants, Poseidon2 hash commitments, and an exact canonical constraint budget of 5,087 R1CS constraints.³ The file `csc.py` in Track A is designated as the canonical constraint-budget authority, and the Circom circuit enforces both the Poseidon2 topology ($t = 9, r = 8$) and the total constraint count as non-negotiable.

Within this design thread, the following additional developments have been articulated and are captured here as defensive publication:

- Clarification of SCN as the learned amortized controller and CSC (“Contraction Spectral Certificate”) as the topology-and-certificate layer binding SCN outputs, governance witnesses, and cryptographic proofs.
- A proposal to unify all drift and protection semantics under a PhaseMirror-HQ memory-kernel layer (FZS-MK with Zeno projection), such that ACE, SCN, and CSC treat this kernel as the sole authority for drift/protection metrics.
- An ADR-based governance and migration plan, including SWOT and RAID analyses, to deprecate duplicate ACE-side drift logic while preserving reproducibility and circuit topology constraints.
- A telemetry contract for exposing kernel-certified scalars (e.g., normalized drift X_n , maximum WAC statistic, a protection index, and a validity bit) into ACE’s certificate and zero-knowledge pipelines, without recomputing semantic logic in the consuming stack.

The final architecture proposed is four-layered:

1. Arithmetic operator layer (cusp-form spaces, Hecke operators, Hermitianization).
2. Kernel authority layer (FZS-MK memory kernel and Zeno projection).
3. Control and certificate layer (SCN, feasibility maps, governance logic, CSC).
4. Cryptographic verification layer (Circom/Rust circuit, Poseidon2 commitments, Groth16 proofs).

This configuration preserves ACE’s original mathematical guarantees while concentrating semantic authority for drift and protection metrics in a single, well-defined kernel.

¹Formal specification in ACE–SCN document. [file:1]

²Definitions of the feasibility map and fidelity modes are given in Sections 4–6 of the ACE–SCN specification. [file:1]

³ACE Roadmap and constraint layout documents. [file:2][file:3]

2 Mathematical Overview of ACE–SCN

2.1 Arithmetic Objects and Hermitianization

Cusp-form space and safe Hecke family. Let

$$V = S_k(\Gamma_0(N))$$

be the complex vector space of weight- k holomorphic cusp forms of level $\Gamma_0(N)$.⁴ For primes $p \nmid N$, the Hecke operators

$$T_p : V \rightarrow V$$

are self-adjoint with respect to the Petersson inner product and form the “safe” part of the Hecke family; non-self-adjoint operators such as U_p for $p \mid N$ are excluded from the base specification.

Computational basis and Gram matrix. Choose a non-diagonal computable basis B for V (for example, a modular-symbols-derived basis), producing a matrix representation

$$M_p$$

for T_p in that basis.⁵ Let $G \succ 0$ denote the Gram matrix of the relevant inner product in basis B . Self-adjointness in this basis is equivalent to the condition

$$M_p G = G M_p^*.$$

A diagnostic self-adjointness error is defined by

$$\text{selfAdjErr}(M_p, G) := \frac{\|M_p G - G M_p^*\|_F}{\|M_p\|_F \|G\|_F}.$$

Stable Hermitianization. Compute a Cholesky factorization $G = LL^*$ and define

$$A_p := L^{-1} M_p L.$$

If $\text{selfAdjErr}(A_p)$ is sufficiently small, then A_p is Hermitian up to numerical tolerance.⁶ Optionally, apply a symmetrization

$$A_p \leftarrow \frac{1}{2}(A_p + A_p^*)$$

and record the residual $\|A_p - A_p^*\|_F$ as a diagnostic.

2.2 Gap Functionals and Perturbation Classes

Eigenvalues and gap functionals. For a Hermitian matrix $A \in \mathbb{C}^{d \times d}$, denote its ordered real eigenvalues by

$$\lambda_1(A) \leq \dots \leq \lambda_d(A).$$

The indexed gap is defined as

$$g_i(A) := \lambda_{i+1}(A) - \lambda_i(A),$$

⁴Section 1 of the ACE–SCN specification. [file:1]

⁵Section 2 of the ACE–SCN specification. [file:1]

⁶Hermitianization and diagnostics described in Section 2. [file:1]

while a smooth soft-min gap functional can be defined as

$$g_\tau(A) := -\tau \log \left(\sum_{i=1}^{d-1} \exp \left(-\frac{\lambda_{i+1}(A) - \lambda_i(A)}{\tau} \right) \right), \quad \tau > 0.$$

These functionals provide smooth objectives for spectral-gap control.⁷

Hecke-span subspace and norms. Fix primes $p_1, \dots, p_r$ with $p_j \nmid N$. Define the Frobenius subspace

$$H_r := \text{span}\{A_{p_1}, \dots, A_{p_r}\} \subset \mathbb{C}^{d \times d}.$$

The Frobenius norm is

$$\|X\|_F := \sqrt{\text{tr}(X^*X)},$$

and the induced distance to H_r is

$$\text{dist}_F(X, H_r) := \min_{H \in H_r} \|X - H\|_F.$$

Fidelity modes and constraint sets. For $\varepsilon > 0$ and $\eta \geq 0$, ACE defines three constraint sets:

$$\begin{aligned} \mathcal{C}_\varepsilon &:= \{\Delta : \Delta = \Delta^*, \|\Delta\|_F \leq \varepsilon\}, \\ \mathcal{C}_\varepsilon^{H_r} &:= \{\Delta : \Delta \in H_r, \Delta = \Delta^*, \|\Delta\|_F \leq \varepsilon\}, \\ \mathcal{C}_{\varepsilon, \eta}^{\text{near}} &:= \{\Delta : \Delta = \Delta^*, \|\Delta\|_F \leq \varepsilon, \text{dist}_F(\Delta, H_r) \leq \eta\}. \end{aligned}$$

These sets correspond respectively to Mode 1 (unstructured), Mode 2 (Hecke-span), and Mode 3 (near-arithmetic) perturbations.⁸

2.3 Projection and Feasibility Map

Frobenius projection onto H_r . Define

$$B = [\text{vec}(A_{p_1}) \cdots \text{vec}(A_{p_r})] \in \mathbb{R}^{d^2 \times r}.$$

Then the Frobenius projection of a matrix X onto H_r is

$$P_{H_r}(X) = \text{unvec}(B(B^\top B)^{-1}B^\top \text{vec}(X)),$$

with the practical recommendation to precompute a numerically stable orthonormal basis for H_r via QR factorization on flattened matrices.⁹

Certificate-first feasibility map. Let Δ_0 be a raw Hermitian proposal (for example, an SCN output after symmetrization). ACE applies a deterministic map F to Δ_0 , depending on the chosen mode.

- Common Hermitianization:

$$\Delta_0 \leftarrow \frac{1}{2}(\Delta_0 + \Delta_0^*).$$

⁷Gap definitions and smooth variants are discussed in Section 3. [file:1]

⁸Fidelity modes appear in Section 4 of the ACE-SCN specification. [file:1]

⁹Implementation notes in Section 5. [file:1]

- Mode 1 (unstructured): scale to the ball of radius ε ,

$$F_1(\Delta_0) := \varepsilon \cdot \frac{\Delta_0}{\max(\|\Delta_0\|_F, \varepsilon)}.$$

- Mode 2 (Hecke-span): project then scale,

$$\begin{aligned} H &= P_{H_r}(\Delta_0), \\ F_2(\Delta_0) &:= \varepsilon \cdot \frac{H}{\max(\|H\|_F, \varepsilon)}. \end{aligned}$$

- Mode 3 (near-arithmetic): clip residual distance and then scale,

$$\begin{aligned} H &= P_{H_r}(\Delta_0), \\ R &= \Delta_0 - H, \\ R' &:= \min\left(1, \frac{\eta}{\|R\|_F}\right) R, \\ \Delta_1 &:= H + R', \\ F_3(\Delta_0) &:= \varepsilon \cdot \frac{\Delta_1}{\max(\|\Delta_1\|_F, \varepsilon)}. \end{aligned}$$

The resulting map $F_m(\Delta_0)$ (for $m \in \{1, 2, 3\}$) is Hermitian and guaranteed to lie in the corresponding constraint set by construction.¹⁰

2.4 Certificates and Key Lemmas

Weyl via Frobenius. For Hermitian A and Hermitian $\Delta \in \mathcal{C}_\varepsilon$, Weyl’s theorem yields the Frobenius-based bound

$$|\lambda_i(A + \Delta) - \lambda_i(A)| \leq \|\Delta\|_2 \leq \|\Delta\|_F \leq \varepsilon.$$

This provides a certified upper bound on eigenvalue movement for all fidelity modes.

Near-span implies near-commuting. If Δ satisfies $\text{dist}_F(\Delta, H_r) \leq \eta$, then for each generator A_{p_i} ,

$$\|[A_{p_i}, \Delta]\|_F \leq 2\|A_{p_i}\|_2 \eta.$$

This inequality turns commutator norms into explicit diagnostics of arithmetic fidelity: small Frobenius distance to the Hecke span implies small commutators.

2.5 SCN Amortized Controller

Dimension-agnostic parametrization. The SCN learns an amortized mapping from instance features to either a raw matrix proposal or a low-dimensional parameter vector (α, β) . A recommended parametrization is

$$\Delta_0(\alpha, \beta) = \sum_{j=1}^r \alpha_j A_{p_j} + \sum_{\ell=1}^s \beta_\ell u_\ell u_\ell^\top,$$

where $u_\ell \in \mathbb{R}^d$ are deterministic pseudorandom unit vectors generated from a fixed seed. This makes the network output dimension independent of d , the ambient matrix size.

¹⁰Feasibility map construction in Section 6. [file:1]

Features and amortized objective. Let $\phi(A, \hat{g})$ be a feature vector including target gap $\hat{g}$, eigenvalue summaries, selected gap values, and quantiles. For a fixed constraint set $\mathcal{C}$, the per-instance constrained problem is

$$\Delta^*(A, \hat{g}) \in \arg \min_{\Delta \in \mathcal{C}} (g_\tau(A + \Delta) - \hat{g})^2.$$

The amortized training objective for SCN parameters θ is

$$\min_{\theta} \mathbb{E}_{(A, \hat{g})} \left[(g_\tau(A + F(\Delta_0(F_\theta(\phi(A, \hat{g})))))) - \hat{g})^2 \right],$$

where F_θ is the network mapping features to parameter vectors, $\Delta_0(\cdot)$ constructs raw perturbations, and $F(\cdot)$ is the deterministic feasibility map.

3 Governance Metrics and Zero-Knowledge Layout

3.1 Track A Governance Metrics

The ACE roadmap introduces a governance stack that computes various drift and retention metrics in Track A.¹¹ Given a proxy distribution μ_{proxy} and a current empirical mean μ_n , define:

- Proxy L_1 norm:

$$\|\mu_{\text{proxy}}\|_1 = \sum_i |\mu_{\text{proxy},i}|.$$

- Support mask and off-support drift:

$$\bar{m}_n = \sum_i |\mu_n(i) - \mu_{\text{proxy}}(i)| (1 - \text{support_mask}_i).$$

- Normalized drift metric:

$$X_n = \frac{\bar{m}_n}{\|\mu_{\text{proxy}}\|_1}.$$

- Retention rate:

$$R_t = \max(0, 1 - \varepsilon - X_n).$$

- Telescoping WAC window metric:

$$W_t = \frac{\bar{m}^{(t+M)}}{\max(\bar{m}^{(t)}, 10^{-12})}.$$

The flag $\text{is_valid} \in \{0, 1\}$ is set based on whether $\max_t W_t$ stays below a prescribed threshold (typically less than 1), serving as a certification gate for windowed contraction.

¹¹Track A description in ACE Roadmap. [file:2]

3.2 Track B Circuit and Constraint Budget

The Track B circuit, implemented in Circom and orchestrated by Rust, is constrained by a canonical 5,087-constraint budget that is locked both in `csc.py` and in a constraint-layout specification.¹² The optimized base circuit uses 133 constraints, decomposed into:

- 64 trajectory consistency constraints:

$$\text{current_mu}[i] \cdot \text{step_n} \equiv \text{h_hat}[i] \pmod{p}.$$

- 64 support-mask booleanity constraints:

$$\text{support_mask}[i](\text{support_mask}[i] - 1) \equiv 0 \pmod{p}.$$

- 1 drift quotient check:

$$X_n \cdot \text{proxy_l1_norm} \equiv \bar{m}_n \cdot \text{SCALE} \pmod{p}.$$

- 1 clamping consistency:

$$R_t \cdot (r_{\text{raw}} - R_t) \equiv 0 \pmod{p}, \quad r_{\text{raw}} = \text{SCALE} - \varepsilon - X_n.$$

- 1 validity-flag booleanity:

$$\text{is_valid}(\text{is_valid} - 1) \equiv 0 \pmod{p}.$$

- 2 topology guards for WAC mode and N_0 consistency.

Poseidon2 hash gadgets (sponge and single-permutation) are then added to reach a projected 4,804 constraints after hashing all witness fields and binding commitments, leaving 283 constraints in reserve for guard-rails and audit hooks.¹³

3.3 Commitment Strategy

Track B uses two Poseidon2 layers:

1. A sponge hash over all relevant witness fields, producing a commitment $h_{\text{commitment}}$.
2. A second Poseidon2 permutation binding the hash and selected outputs (e.g., X_n , R_t , is_valid , WAC product, and a retry nonce) into a final CAS commitment.

In Circom-style pseudocode:

```
component poseidon_h = Poseidon2Sponge(205, 9, 8);
for (var i = 0; i < 205; i++) {
    poseidon_h.in[i] <== witness_field[i];
}
signal h_commitment;
h_commitment <== poseidon_h.out;

component poseidon_gamma = Poseidon2(5, 9, 8);
```

¹²Constraint layout document. [file:3]

¹³Poseidon2 integration roadmap and budget in constraint layout. [file:3]


```

poseidon_gamma.in[0] <== h_commitment;
poseidon_gamma.in[1] <== X_n;
poseidon_gamma.in[2] <== R_t;
poseidon_gamma.in[3] <== max_wac_product;
poseidon_gamma.in[4] <== retry_nonce;
cas_commitment <== poseidon_gamma.out;

```

This structure is tightly coupled to the 5,087-constraint budget and enforces both determinism and collision resistance.¹⁴

4 PhaseMirror-HQ Unification and Kernel Authority

4.1 Motivation for Single Semantic Authority

In the current ACE design, Track A defines the semantics of drift and protection metrics (e.g. X_n , WAC, and `is_valid`), while Track B only verifies algebraic consistency and commitments. This leaves open the possibility of duplicating or partially re-implementing drift logic in other systems interacting with ACE.

The proposed unification introduces a dedicated *kernel authority layer*, implemented by PhaseMirror-HQ, whose FZS-MK memory kernel and Zeno projection define all drift and protection metrics consumed by ACE, SCN, and CSC. Under this model:

- Semantic definitions of drift and protection reside entirely in the kernel layer.
- ACE and SCN treat these metrics as read-only telemetry.
- CSC verifies that these telemetry values are properly bound into commitments and respect topological and budget constraints, but does not define their semantics.

4.2 Kernel Telemetry Contract

A minimal telemetry record emitted by PhaseMirror could be

$$\Theta_{\text{kernel}} = \left(X_n^{\text{kernel}}, W_{\text{max}}^{\text{kernel}}, \zeta_{\text{protect}}, \text{is_valid}_{\text{kernel}} \right),$$

augmented with a version field for schema evolution. ACE would then consume this record as control telemetry and include its fields in certificate and CAS commitments.

A Python sketch of this interface is:

```

from dataclasses import dataclass

@dataclass(frozen=True)
class KernelTelemetry:
    xn_kernel: float
    wt_max_kernel: float
    protection_zeta: float
    is_valid_kernel: bool
    telemetry_version: int = 1

def load_kernel_telemetry(step_state) -> KernelTelemetry:
    # Delegates semantic authority to PhaseMirror-HQ/kernel/core/zeno
    return zeno.compute_kernel_telemetry(step_state)

```

¹⁴Detailed in the constraint layout and roadmap. [file:2][file:3]

An ACE-side wrapper can optionally maintain a *parity mode* during migration:

```
def get_drift_metrics(step_state, parity_mode: bool = False):
    kt = load_kernel_telemetry(step_state)
    if parity_mode:
        legacy_xn = compute_legacy_xn(step_state)
        assert abs(legacy_xn - kt.xn_kernel) < 1e-9
    return kt
```

4.3 SCN Conditioning on Kernel Telemetry

With kernel authority established, the SCN feature map is extended from $\phi(A, \hat{g})$ to

$$\phi(A, \hat{g}, \Theta_{\text{kernel}}),$$

allowing the controller to react to kernel-defined drift and protection states without recomputing or redefining them. The feasibility map and spectral-control objective remain unchanged; only the policy distribution learned by SCN is conditioned on the additional telemetry.

5 ADR, SWOT, and RAID Structure

5.1 ADR Package

An ADR (Architecture Decision Record) package for this unification can be organized as:

- **ADR-001:** FZS-MK/Zeno as Single Drift and Protection Authority.
- **ADR-002:** ACE-PhaseMirror Kernel Telemetry Contract.
- **ADR-003:** SCN/CSC Conditioning on Kernel Metrics Only.

Each ADR should include context, decision, rationale, alternatives, consequences, SWOT, and a RAID log. A skeletal structure (in prose form) is:

ADR-001: FZS-MK/Zeno as Single Drift and Protection Authority

- **Context:** Drift and protection metrics currently defined in ACE Track A; PhaseMirror kernel provides a potentially richer and more general non-Markovian framework.
- **Decision:** Declare PhaseMirror’s FZS-MK/Zeno layer the sole semantic authority for drift and protection metrics used by ACE, SCN, and CSC.
- **Consequences:** ACE may consume and certify these metrics but must not define independent semantic formulas after migration completes.

Similar detail can be elaborated for ADR-002 and ADR-003, focusing on schema-level contracts and SCN feature constraints, respectively.

5.2 SWOT Analysis

Strengths.

- ACE already separates learned proposal generation from deterministic feasibility enforcement, making authority unification structurally compatible.¹⁵
- The Track A / Track B split ensures that semantic computation resides outside the circuit, simplifying kernel integration.¹⁶
- A fixed 5,087-constraint budget with Poseidon2 topology lock yields strong reproducibility and auditability.¹⁷

Weaknesses.

- Current documentation does not yet specify the exact ACE–PhaseMirror API boundary.
- Moving semantics to PhaseMirror increases coupling to kernel release cycles and failure modes.
- The circuit relies on Track A for semantic correctness, so kernel bugs are not directly ruled out by the R1CS constraints.¹⁸

Opportunities.

- Prevents semantic drift across training, runtime, and proof-generation pipelines.
- Telemetry contract enables both open-core and OEM deployment paths without altering high-level mathematics.¹⁹
- Reserved circuit headroom (283 constraints) allows introducing future guard rails or audit hooks around kernel telemetry if needed.²⁰

Threats.

- Divergent telemetry schemas between ACE and PhaseMirror could break determinism or governance clarity.
- Overly complex telemetry could strain Poseidon2 input layout or circuit budget.
- Partial or inconsistent deprecation of ACE-side drift logic could undermine the single-authority guarantee.

5.3 RAID Log (Risks, Assumptions, Issues, Dependencies)

Risks.

- **R1:** Numerical divergence between kernel and legacy ACE drift metrics during migration.
- **R2:** Telemetry schema churn compromising certificate determinism.
- **R3:** Telemetry additions inadvertently increasing constraint count beyond budget.

¹⁵SCN and feasibility map design. [file:1]

¹⁶Roadmap and constraint layout. [file:2][file:3]

¹⁷Constraint layout. [file:3]

¹⁸Constraint responsibilities. [file:3]

¹⁹Deployment discussion in roadmap. [file:2]

²⁰Constraint budget headroom. [file:3]

Assumptions.

- **A1:** PhaseMirror kernel can produce stable, scalar (or low-dimensional) telemetry suitable for ACE.
- **A2:** Circuit headroom is sufficient for any minor witness-binding refinements.²¹

Issues.

- **I1:** Formalizing a versioned KernelTelemetry schema.
- **I2:** Establishing a deprecation policy and test coverage for legacy ACE drift/WAC logic.

Dependencies.

- **D1:** PhaseMirror-HQ kernel API release.
- **D2:** ACE certificate schema updates and Track B witness adapter changes.

6 Code Snippets and Implementation Sketches

6.1 Python: Kernel Telemetry and Certificate Payload

```
from dataclasses import dataclass

@dataclass(frozen=True)
class KernelTelemetry:
    xn_kernel: float
    wt_max_kernel: float
    protection_zeta: float
    is_valid_kernel: bool
    telemetry_version: int = 1

def load_kernel_telemetry(step_state) -> KernelTelemetry:
    # Delegates semantic authority to PhaseMirror-HQ/kernel/core/zeno
    return zeno.compute_kernel_telemetry(step_state)

def get_drift_metrics(step_state, parity_mode: bool = False):
    kt = load_kernel_telemetry(step_state)
    if parity_mode:
        legacy_xn = compute_legacy_xn(step_state)
        assert abs(legacy_xn - kt.xn_kernel) < 1e-9
    return kt

def certificate_payload(theta_base, telemetry, outputs):
    return {
        "theta": serialize_theta(theta_base),
        "xn_kernel": telemetry.xn_kernel,
        "wt_max_kernel": telemetry.wt_max_kernel,
        "protection_zeta": telemetry.protection_zeta,
        "is_valid_kernel": int(telemetry.is_valid_kernel),
```

²¹Headroom explicitly reserved for future guard-rails. [file:3]

```

    "telemetry_version": telemetry.telemetry_version,
    "outputs": outputs,
}

```

6.2 Circom Witness Binding Sketch

Adapting the existing CAS commitment pattern, a telemetry-aware binding might be:

```

component poseidon_gamma = Poseidon2(5, 9, 8);
poseidon_gamma.in[0] <== h_commitment;
poseidon_gamma.in[1] <== xn_kernel;
poseidon_gamma.in[2] <== retention_rate;
poseidon_gamma.in[3] <== max_wac_product;
poseidon_gamma.in[4] <== retry_nonce;
cas_commitment <== poseidon_gamma.out;

```

Here, `xn_kernel` replaces or augments the previous X_n witness field while preserving circuit budget and topology constraints.

7 Validation Path and Repository Changes

7.1 Four-Phase Validation Plan

A fast yet robust validation path is:

1. **Authority Declaration:** Accept ADR-001 and update documentation to state that PhaseMirror’s kernel layer is the semantic authority for drift and protection metrics.
2. **Parity Phase:** Run existing ACE drift/WAC logic in parallel with kernel telemetry on benchmark datasets and assert numerical agreement within strict tolerances.
3. **Contract Phase:** Integrate `KernelTelemetry` fields into ACE certificates and `Poseidon2` witness commitments, keeping SCN behavior unchanged initially.
4. **Control Phase:** Retrain or fine-tune SCN to condition on kernel telemetry, then compare to pre-integration baselines in terms of objective values, fidelity metrics, proof determinism, and runtime.

7.2 Recommended Repository Edits

Within the ACE repository structure outlined in the roadmap,²² recommended edits include:

- **src/ace/governor.py:** Replace internal ownership of X_n with calls into `get_drift_metrics`.
- **src/ace/monitor.py:** Route WAC-related decisions via kernel telemetry.
- **src/ace/certify.py:** Extend the certificate schema with telemetry fields and versioning.
- **src/ace/cas_registry.py:** Ensure telemetry fields are hashed consistently into CAS commitments.

²²ACE Core scaffold and modules. [file:2]

- **src/ace/csc.py** and Circom circuit: Maintain the 5,087-constraint lock and adjust witness binding if necessary within the existing budget.
- **tests/**: Add parity tests (kernel vs legacy drift), schema-version tests, and checks that forbid reintroduction of duplicate drift logic.
- **adr/**: Add ADR-001, ADR-002, and ADR-003, each with its own embedded SWOT and RAID sections.

8 Defensive Publication Notes

This report is intended as a defensive disclosure of the integrated ACE-SCN-CSC-PhaseMirror architecture. It documents:

- The mathematical specification of ACE-SCN, including fidelity modes, feasibility maps, and certified commutator bounds.
- The Track A / Track B split and the fixed 5,087-constraint Circom topology with Poseidon2 commitments.
- The conceptual role of a Contraction Spectral Certificate as a certificate-and-topology layer binding controller outputs to cryptographic proofs.
- The unification with a memory-kernel and projection-based authority layer for drift and protection metrics.
- Concrete code and interface sketches for telemetry integration, ADR governance structures, SWOT and RAID analyses, and validation pathways.

For maximum defensive value, it is recommended to:

- Publish this document (or an equivalent) in a publicly timestamped venue or repository.
- Include immutable hashes or content-addressable identifiers for this text and associated ADRs.
- Maintain accompanying machine-readable schemas (e.g., JSON schemas for `KernelTelemetry` and `ACECertificate`) in a public code repository.

9 Conclusion

The developments synthesized here refine rather than replace the ACE program. ACE remains a certified spectral-control engine on arithmetic operators; SCN remains the amortized proposal mechanism; CSC remains the topology-and-certificate authority; and a PhaseMirror FZS-MK/Zeno layer becomes the single semantic authority for drift and protection metrics. This layered arrangement is mathematically coherent, operationally auditable, and appropriate for a prior-art defensive publication.

A Mathematical Appendix: Proofs and Operator Norm Bounds

This appendix collects explicit proofs and operator-norm bounds underlying the ACE–SCN specification, with emphasis on feasibility-map invariants, spectral perturbation bounds, and commutator estimates used as arithmetic-fidelity diagnostics.²³ Throughout, $\|\cdot\|_2$ denotes the operator norm induced by the Euclidean norm on $\mathbb{C}^d$, and $\|\cdot\|_F$ denotes the Frobenius norm.

A.1 Weyl-Type Bounds via Frobenius Norm

ACE uses Weyl’s inequality together with the relation $\|\cdot\|_2 \leq \|\cdot\|_F$ to certify that any perturbation in $\mathcal{C}_\varepsilon$ moves eigenvalues by at most ε in absolute value.²⁴

[Weyl via Frobenius] Let $A \in \mathbb{C}^{d \times d}$ be Hermitian and let $\Delta \in \mathbb{C}^{d \times d}$ be Hermitian with $\|\Delta\|_F \leq \varepsilon$. Denote the ordered eigenvalues of A and $A + \Delta$ by

$$\lambda_1(A) \leq \dots \leq \lambda_d(A), \quad \lambda_1(A + \Delta) \leq \dots \leq \lambda_d(A + \Delta).$$

Then for each $i \in \{1, \dots, d\}$,

$$|\lambda_i(A + \Delta) - \lambda_i(A)| \leq \|\Delta\|_2 \leq \|\Delta\|_F \leq \varepsilon.$$

Proof. The first inequality is Weyl’s eigenvalue perturbation bound for Hermitian matrices:

$$|\lambda_i(A + \Delta) - \lambda_i(A)| \leq \|\Delta\|_2, \quad i = 1, \dots, d.$$

This is a standard result; see, e.g., classical matrix analysis texts. The second inequality

$$\|\Delta\|_2 \leq \|\Delta\|_F$$

follows because the Frobenius norm is the ℓ_2 -norm of the singular values, and the operator norm is the ℓ_∞ -norm of the same singular values, so $\|\cdot\|_2 \leq \|\cdot\|_F$ always holds. The final inequality is by the assumption $\|\Delta\|_F \leq \varepsilon$, completing the chain. $\square$

As an immediate corollary, any perturbation produced by the feasibility map in Mode 1 (which outputs an element of $\mathcal{C}_\varepsilon$) moves every eigenvalue of the target operator A by at most ε .

A.2 Feasibility-Map Invariants in All Modes

We next give a unified proof that the feasibility maps F_1, F_2, F_3 used in ACE preserve Hermitian symmetry and enforce the appropriate mode constraints.²⁵

[Feasibility-map invariants] Let $\Delta_0 \in \mathbb{C}^{d \times d}$ be arbitrary, and let

$$\Delta'_0 := \frac{1}{2}(\Delta_0 + \Delta_0^*)$$

²³See Sections 4–9 of the ACE–SCN formal specification for the informal statements. [file:1]

²⁴The set $\mathcal{C}_\varepsilon$ is defined in Section 4.3 of the ACE–SCN document. [file:1]

²⁵See Section 6 of the ACE–SCN specification for the construction of F_m . [file:1]

be its Hermitian part. Let F_1, F_2, F_3 be defined as in the ACE specification, using Δ'_0 as input:

$$\begin{aligned}
F_1(\Delta'_0) &:= \varepsilon \cdot \frac{\Delta'_0}{\max(\|\Delta'_0\|_F, \varepsilon)}, \\
H(\Delta'_0) &:= P_{H_r}(\Delta'_0), \\
F_2(\Delta'_0) &:= \varepsilon \cdot \frac{H(\Delta'_0)}{\max(\|H(\Delta'_0)\|_F, \varepsilon)}, \\
R(\Delta'_0) &:= \Delta'_0 - H(\Delta'_0), \\
R'(\Delta'_0) &:= \min\left(1, \frac{\eta}{\|R(\Delta'_0)\|_F}\right) R(\Delta'_0), \\
\Delta_1(\Delta'_0) &:= H(\Delta'_0) + R'(\Delta'_0), \\
F_3(\Delta'_0) &:= \varepsilon \cdot \frac{\Delta_1(\Delta'_0)}{\max(\|\Delta_1(\Delta'_0)\|_F, \varepsilon)}.
\end{aligned}$$

Then:

- (a) For each $m \in \{1, 2, 3\}$, $F_m(\Delta'_0)$ is Hermitian.
- (b) $F_1(\Delta'_0) \in \mathcal{C}_\varepsilon$.
- (c) $F_2(\Delta'_0) \in \mathcal{C}_\varepsilon^{H_r}$.
- (d) $F_3(\Delta'_0) \in \mathcal{C}_{\varepsilon, \eta}^{\text{near}}$.

Proof. (a) Hermitian symmetry is preserved in each case because all operations used in the construction are Hermitian-preserving:

- Δ'_0 is Hermitian by definition.
- The Frobenius projection P_{H_r} onto a linear subspace H_r preserves Hermitian structure when H_r is spanned by Hermitian matrices A_{p_j} . That is, if $\Delta'_0 = \Delta'^{*}_0$, then $H(\Delta'_0)$ is a real linear combination of the A_{p_j} , hence Hermitian.
- The residual $R(\Delta'_0) = \Delta'_0 - H(\Delta'_0)$ is the difference of Hermitian matrices, hence Hermitian.
- Scaling by a real scalar preserves Hermitian structure, so $R'(\Delta'_0)$ and the normalized outputs $F_m(\Delta'_0)$ remain Hermitian.

(b) For Mode 1, define

$$\Delta^{(1)} := F_1(\Delta'_0) = \varepsilon \cdot \frac{\Delta'_0}{\max(\|\Delta'_0\|_F, \varepsilon)}.$$

By construction, $\Delta^{(1)} = (\Delta^{(1)})^*$. Its Frobenius norm is

$$\|\Delta^{(1)}\|_F = \varepsilon \cdot \frac{\|\Delta'_0\|_F}{\max(\|\Delta'_0\|_F, \varepsilon)} \leq \varepsilon,$$

since the fraction is at most 1. Thus $\Delta^{(1)} \in \mathcal{C}_\varepsilon$.

(c) For Mode 2, define

$$\Delta^{(2)} := F_2(\Delta'_0) = \varepsilon \cdot \frac{H(\Delta'_0)}{\max(\|H(\Delta'_0)\|_F, \varepsilon)}.$$

We have $H(\Delta'_0) \in H_r$ by definition of the projection. As above, the scaling factor ensures

$$\|\Delta^{(2)}\|_F \leq \varepsilon.$$

Hermitianity of $H(\Delta'_0)$ was established in part (a). Therefore, $\Delta^{(2)} \in \mathcal{C}_\varepsilon^{H_r}$.

(d) For Mode 3, write

$$\Delta^{(3)} := F_3(\Delta'_0) = \varepsilon \cdot \frac{\Delta_1(\Delta'_0)}{\max(\|\Delta_1(\Delta'_0)\|_F, \varepsilon)}.$$

By part (a), $\Delta_1(\Delta'_0)$ is Hermitian. The final scaling ensures

$$\|\Delta^{(3)}\|_F \leq \varepsilon.$$

It remains to show that $\text{dist}_F(\Delta^{(3)}, H_r) \leq \eta$. Let

$$\tilde{\Delta} := \Delta_1(\Delta'_0) = H(\Delta'_0) + R'(\Delta'_0).$$

We have $H(\Delta'_0) \in H_r$ and $R'(\Delta'_0)$ orthogonal (in the Frobenius geometry) to H_r because it is a scaled version of the residual from the projection. Thus

$$\text{dist}_F(\tilde{\Delta}, H_r) = \|R'(\Delta'_0)\|_F.$$

By definition of $R'(\Delta'_0)$,

$$\|R'(\Delta'_0)\|_F = \min\left(1, \frac{\eta}{\|R(\Delta'_0)\|_F}\right) \|R(\Delta'_0)\|_F \leq \eta.$$

Now note that the final scaling

$$\Delta^{(3)} = \varepsilon \cdot \tilde{\Delta} / \max(\|\tilde{\Delta}\|_F, \varepsilon)$$

shrinks $\tilde{\Delta}$ (or leaves it unchanged) by a nonnegative scalar factor no larger than 1. In particular, distances to H_r (measured in Frobenius norm) can only decrease under such a contraction:

$$\text{dist}_F(\Delta^{(3)}, H_r) \leq \text{dist}_F(\tilde{\Delta}, H_r) = \|R'(\Delta'_0)\|_F \leq \eta.$$

Therefore, $\Delta^{(3)} \in \mathcal{C}_{\varepsilon, \eta}^{\text{near}}$, completing the proof. $\square$

A.3 Near-Span Implies Near-Commuting

ACE interprets commutator norms $\|[A_{p_i}, \Delta]\|_F$ as diagnostics for arithmetic fidelity in Mode 3.²⁶ Here we spell out the standard operator-norm argument that justifies the bound.

[Near-span implies near-commuting] Let $A_{p_i} \in \mathbb{C}^{d \times d}$ be Hermitian operators (e.g. the safe Hecke matrices) spanning H_r , and let

$$\Delta \in \mathcal{C}_{\varepsilon, \eta}^{\text{near}}$$

so that $\Delta = \Delta^*$ and $\text{dist}_F(\Delta, H_r) \leq \eta$. Then for each generator A_{p_i} ,

$$\|[A_{p_i}, \Delta]\|_F \leq 2 \|A_{p_i}\|_2 \eta.$$

²⁶See Section 9.2 of the ACE-SCN specification. [file:1]

Proof. By definition of the distance, there exists $H \in H_r$ such that

$$\|\Delta - H\|_F = \text{dist}_F(\Delta, H_r) \leq \eta.$$

We can decompose

$$[A_{p_i}, \Delta] = [A_{p_i}, H] + [A_{p_i}, \Delta - H].$$

Since $H \in H_r$ and H_r is spanned by the commuting family of safe Hecke operators A_{p_j} , H commutes with each A_{p_i} , giving

$$[A_{p_i}, H] = 0.$$

Therefore,

$$[A_{p_i}, \Delta] = [A_{p_i}, \Delta - H] = A_{p_i}(\Delta - H) - (\Delta - H)A_{p_i}.$$

Using submultiplicativity of the operator and Frobenius norms,

$$\begin{aligned} \|[A_{p_i}, \Delta]\|_F &= \|A_{p_i}(\Delta - H) - (\Delta - H)A_{p_i}\|_F \\ &\leq \|A_{p_i}(\Delta - H)\|_F + \|(\Delta - H)A_{p_i}\|_F \\ &\leq \|A_{p_i}\|_2 \|\Delta - H\|_F + \|\Delta - H\|_F \|A_{p_i}\|_2 \\ &= 2 \|A_{p_i}\|_2 \|\Delta - H\|_F \\ &\leq 2 \|A_{p_i}\|_2 \eta, \end{aligned}$$

as claimed. $\square$

This proposition is the precise content behind ACE's statement that Mode 3 commutator norms scale as $O(\eta)$ and can be interpreted as certified arithmetic-fidelity diagnostics.

A.4 Monotonicity and Scaling Behaviour in Mode 3

ACE informally notes that in Mode 3, smaller η produces stricter control of arithmetic fidelity and smaller commutators while Mode 2 corresponds to the endpoint $\eta = 0$.²⁷ We briefly formalize these monotonicity properties.

[Monotonicity in η] Fix $\varepsilon > 0$, and consider two near-arithmetic constraint sets

$$\mathcal{C}_{\varepsilon, \eta_1}^{\text{near}} \subseteq \mathcal{C}_{\varepsilon, \eta_2}^{\text{near}} \quad \text{with} \quad 0 \leq \eta_1 \leq \eta_2.$$

Then the set of admissible perturbations under η_1 is contained in the admissible set under η_2 . In particular, any admissible Δ for η_1 is also admissible for η_2 , and the corresponding commutator bounds satisfy

$$\|[A_{p_i}, \Delta]\|_F \leq 2 \|A_{p_i}\|_2 \eta_1 \quad \Rightarrow \quad \|[A_{p_i}, \Delta]\|_F \leq 2 \|A_{p_i}\|_2 \eta_2.$$

Proof. The inclusion of sets is immediate from the definition:

$$\mathcal{C}_{\varepsilon, \eta_1}^{\text{near}} = \{\Delta : \Delta = \Delta^*, \|\Delta\|_F \leq \varepsilon, \text{dist}_F(\Delta, H_r) \leq \eta_1\},$$

and if $\eta_1 \leq \eta_2$, then every Δ satisfying $\text{dist}_F(\Delta, H_r) \leq \eta_1$ also satisfies $\text{dist}_F(\Delta, H_r) \leq \eta_2$. The rest follows from Proposition A.3. $\square$

²⁷See Section 12 of the ACE-SCN specification for the qualitative description of expected outcomes. [file:1]

[Mode 2 as endpoint] Mode 2 corresponds exactly to the case $\eta = 0$ in Mode 3. In this limit, the near-arithmetic constraint set reduces to the Hecke-span constraint set, and the commutator bound becomes

$$\|[A_{p_i}, \Delta]\|_F = 0$$

for all A_{p_i} , since $\Delta \in H_r$ commutes with each generator.

Proof. If $\eta = 0$, then $\text{dist}_F(\Delta, H_r) \leq 0$ implies $\Delta \in H_r$, so

$$\mathcal{C}_{\varepsilon,0}^{\text{near}} = \mathcal{C}_{\varepsilon}^{H_r}.$$

The commutator statement follows as in Proposition A.3 with $\eta = 0$. $\square$

A.5 Operator Norm Bounds in the Drift Quotient Constraint

The constraint-layout document encodes the normalized drift metric X_n and the off-support drift sum $\bar{m}_n$ via a single multiplicative constraint

$$X_n \cdot \text{proxy_l1_norm} \equiv \bar{m}_n \cdot \text{SCALE} \pmod{p},$$

which, in real arithmetic, corresponds to

$$X_n = \frac{\bar{m}_n}{\|\mu_{\text{proxy}}\|_1}.$$

28

While this is an exact identity rather than an inequality, it interacts with the norm bounds above via the role of X_n in retention and feasibility constraints. For completeness, we note the following simple bound.

[Drift and retention bounds] Assume the proxy distribution μ_{proxy} is normalized so that $\|\mu_{\text{proxy}}\|_1 > 0$, and let

$$\bar{m}_n = \sum_i |\mu_n(i) - \mu_{\text{proxy}}(i)| (1 - \text{support_mask}_i)$$

with $\text{support_mask}_i \in \{0, 1\}$. Then

$$0 \leq X_n = \frac{\bar{m}_n}{\|\mu_{\text{proxy}}\|_1}$$

and the retention rate

$$R_t = \max(0, 1 - \varepsilon - X_n)$$

satisfies

$$0 \leq R_t \leq 1.$$

Proof. By definition, each summand in $\bar{m}_n$ is nonnegative, hence $\bar{m}_n \geq 0$. With $\|\mu_{\text{proxy}}\|_1 > 0$, this implies $X_n \geq 0$. The upper bound $R_t \leq 1$ follows from

$$R_t = \max(0, 1 - \varepsilon - X_n) \leq 1,$$

since $\varepsilon > 0$ and $X_n \geq 0$. Nonnegativity is immediate from the max operation. $\square$

This simple bound is one of the ingredients justifying the clamping constraint

$$R_t(r_{\text{raw}} - R_t) = 0, \quad r_{\text{raw}} = \text{SCALE} - \varepsilon - X_n$$

in the circuit, which encodes $R_t = \max(0, r_{\text{raw}})$ in arithmetic form.²⁹

²⁸Normalization constraint as in the ACE constraint layout. [file:3]

²⁹Clamping constraint described in the constraint layout. [file:3]

A.6 Summary

The bounds and proofs in this appendix make explicit the operator-norm control that underpins ACE’s spectral guarantees:

- Feasibility maps yield perturbations lying in the correct constraint sets while preserving Hermiticity.
- Eigenvalue movements are bounded by the Frobenius radius ε via Weyl’s inequality.
- Commutator norms behave as $O(\eta)$ in Mode 3, with Mode 2 as the commuting endpoint.
- Drift and retention metrics used in the Track A governance stack respect natural range constraints that are encoded succinctly in the Track B circuit.

These explicit derivations complement the main specification and strengthen the correctness claims made in the ACE–SCN and constraint-layout documents.³⁰

References

- [1] Michael Reed and Barry Simon. *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*. Academic Press, New York, 1980. Classical reference for self-adjoint operators, spectral theorem, and operator norm properties.
- [2] Tosio Kato. *Perturbation Theory for Linear Operators*. Classics in Mathematics. Springer, Berlin, reprint of the 1980 edition edition, 1995. Canonical source for Weyl inequalities, perturbation bounds, and spectral stability results.
- [3] Rajendra Bhatia. *Matrix Analysis*. Springer, New York, 1997. Standard reference for matrix norms, operator norms, commutators, and inequality techniques.
- [4] Walter Rudin. *Functional Analysis*. McGraw–Hill, New York, 1991. Foundations of Hilbert spaces, bounded linear operators, self-adjointness, and spectral theory.
- [5] John B. Conway. *A Course in Functional Analysis*. Springer, New York, 1990. Hilbert space operators, adjoints, operator norms, and basic spectral theory.
- [6] Wikipedia contributors. Self-adjoint operator. https://en.wikipedia.org/wiki/Self-adjoint_operator, 2026. Background summary on self-adjoint operators and real spectrum properties.
- [7] Christopher Heil. Adjoints in hilbert spaces. https://heil.math.gatech.edu/handouts/adjoint_hilbert.pdf, 2007. Lecture notes covering bounded self-adjoint operators and norms in Hilbert spaces.
- [8] E. Brian Davies. *Linear Operators and Their Spectra*, volume 106 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2007. Advanced spectral theory for operators on Hilbert spaces; useful for spectral-gap discussions.

³⁰See the ACE formal specification and constraint layout for the high-level statements this appendix formalizes. [file:1][file:3]

- [9] Delin Yang. On the norm of a self-adjoint operator and applications to bilinear inequalities. *Bulletin of the Belgian Mathematical Society — Simon Stevin*, 3(2):177–190, 1996. Discusses characterization and bounds for norms of self-adjoint operators.
- [10] Bernd Thaller. *Self-Adjoint Operators in Quantum Mechanics*, volume 14 of *SMF/AMS Texts and Monographs*. American Mathematical Society, Providence, RI, 2000. Detailed treatment of self-adjointness, spectra, and operator norms in a quantum-mechanical context.
- [11] Emmanuel Kowalski. Spectral theory in hilbert spaces. <https://people.math.ethz.ch/~kowalski/spectral-theory.pdf>, 2009. Lecture notes on spectral theory for families of unitary/self-adjoint operators, relevant for Hecke operators.
- [12] Joel A. Tropp, Stanislav Minsker, and coauthors. Intrinsic dimension concentration inequalities for self-adjoint operators. *arXiv preprint*, arXiv:2602.13465, 2026. Provides concentration inequalities for sums of self-adjoint operators and operator-norm bounds.
- [13] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 2 edition, 2013. Comprehensive reference for eigenvalue perturbation, Weyl inequalities, and commutator estimates.
- [14] Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, Cambridge, 1991. Supplementary material on matrix inequalities, operator norms, and spectral bounds.
- [15] Terence Tao. *Topics in Random Matrix Theory*, volume 132 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2012. Useful for intuition on spectral gaps, eigenvalue spacing, and perturbation in high dimensions.
- [16] Florine Hartwig, Josua Sassen, Omri Azencot, Martin Rumpf, and Mirela Ben-Chen. An elastic basis for spectral shape correspondence. 2023. Example of spectral methods using non-orthogonal bases and functional maps; conceptually adjacent to spectral shaping.